# Scaling Spherical CNNs

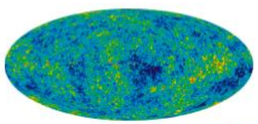
Carlos Esteves Jean-Jacques Slotine Ameesh Makadia
machc@google.com jjs@mit.edu makadia@google.com

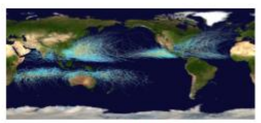
github.com/google-research/spherical-cnn

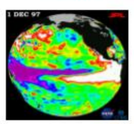
## Introduction

### Spherical CNNs

- Principled way of learning from spherical signals, using spherical convolutions computed spectrally.
- Equivariant to 3D rotations.
- Respects spherical topology, robust to sampling.


Cosmic radiation (NASA / WMAP Science Team)


Cyclone tracks (NASA, Nilfanion)


Sea surface temperature

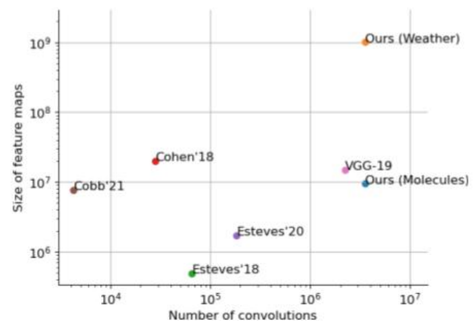
Spherical panorama (SUN360)

Molecules

3D CAD models (Modelnet40)

### Contributions

- Spherical CNNs 100x larger than previously possible.
- More expressive and efficient layers and activations.
- Application-specific modeling for molecules and weather.



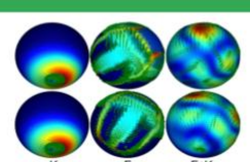

### Related work

Cohen et al (ICLR'18), Esteves et al (ECCV'18, NeurIPS'20), Kondor et al (NeurIPS'18), Cobb et al (ICLR'21), McEwen et al (ICLR'22), Ocampo et al (ICLR'23).

## Method

- Based on spin-weighted SCNNs (Esteves et al, NeurIPS'20).
- Phase collapse activation: $x_0 \leftarrow W_1 x_0 + W_2 |x| + b$
  - Inspired by Guth et al, arXiv'21.
- Efficient spin-weighted spherical harmonics on TPUs
  - Avoid tensor slicing/concatenation.
  - Use DFT matrices instead of FFT.
- Spectral batch normalization, pooling and residual connection.

Spectral residual block

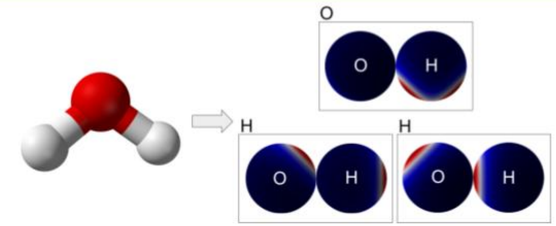


Spin-weighted spherical convolution with spin=1 as a vector field

| | Δ Steps/s [%] ↑ | Δ RMSE [%] ↓ |
|---|---|---|
| JAX implementation | 33.7 | 0.0 |
| Phase collapse | −4.6 | −8.0 |
| No Δ symmetries | 16.3 | 0.0 |
| Use DFT | 21.4 | 0.0 |
| Spectral batch norm | 7.8 | −1.4 |
| Efficient residual | 19.3 | −2.4 |

## Molecular property regression

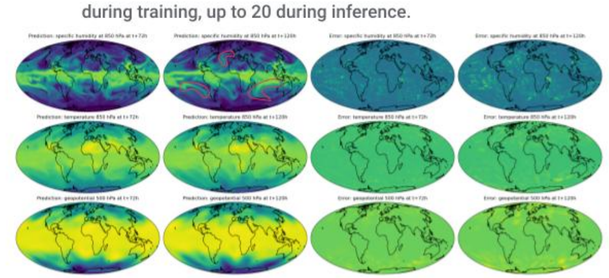
- Represent a molecule as a set of spheres per atom, regress a set of properties.
- One feature channel per atom type.
- Use directional attributes only (no radius).
- $f_{iz}(x) = \sum_{z_j = z} \frac{z_i z_j}{|r_{ij}|^p} e^{-\frac{(x \cdot r_{ij})^2}{\beta |r_{ij}|}}$




| | | $\mu$ [$D$] | $\alpha$ [$a_0^3$] | $\epsilon_{\text{HOMO}}$ [meV] | $\epsilon_{\text{LUMO}}$ [meV] | $\epsilon_{\text{gap}}$ [meV] | $< R^2 >$ [$a_0^2$] | zpve [meV] | $U_0$ [meV] | U [meV] | H [meV] | G [meV] | $C_v$ [$\frac{\text{cal}}{\text{mol K}}$] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Split 1 | DimeNet++ (2020) | 0.030 | 0.044 | 24.6 | 19.5 | 32.6 | 0.331 | 1.21 | 6.32 | 6.28 | 6.53 | 7.56 | 0.023 |
| | PaiNN (2021) | 0.012 | 0.045 | 27.6 | 20.4 | 45.7 | 0.066 | 1.28 | 5.85 | 5.83 | 5.98 | 7.35 | 0.024 |
| | TorchMD-Net (2022) | 0.011 | 0.059 | 20.3 | 17.5 | 36.1 | 0.033 | 1.84 | 6.15 | 6.38 | 6.16 | 7.62 | 0.026 |
| | Ours | 0.016 | 0.049 | 21.6 | 18.0 | 28.8 | 0.027 | 1.15 | 5.65 | 5.72 | 5.69 | 6.54 | 0.022 |
| Split 2 | EGNN (2021) | 0.029 | 0.071 | 29.0 | 25.0 | 48.0 | 0.106 | 1.55 | 11.00 | 12.00 | 12.00 | 12.00 | 0.031 |
| | SEGNN (2022) | 0.023 | 0.060 | 24.0 | 21.0 | 42.0 | 0.660 | 1.62 | 15.00 | 13.00 | 16.00 | 15.00 | 0.031 |
| | Equiformer (2022) | 0.014 | 0.056 | 17.0 | 16.0 | 33.0 | 0.227 | 1.32 | 10.00 | 11.00 | 10.00 | 10.00 | 0.025 |
| | Ours | 0.017 | 0.049 | 22.3 | 19.1 | 29.8 | 0.028 | 1.19 | 5.96 | 5.98 | 5.97 | 6.97 | 0.023 |

## Weather forecasting

- Take a set of atmospheric variables like temperature, wind, humidity, and predict their future values with fully convolutional models.
- WeatherBench (Rasp et al): predict directly 3 or 5 days ahead.
- Keisler (arXiv'22): predict iteratively in 6h increments.
  - Our largest model: 256x256x78 inputs and outputs; iterated for 12 steps during training, up to 20 during inference.



| | 1 day | | 3 days | | 5 days | |
|---|---|---|---|---|---|---|
| | Z500 | T850 | Z500 | T850 | Z500 | T850 |
| Keisler (2022) | 64.9 | 0.730 | 175.5 | 1.17 | 344.7 | 1.78 |
| Ours | 58.3 | 0.827 | 167.2 | 1.26 | 340.0 | 1.91 |

| | 3 days | | | 5 days | | |
|---|---|---|---|---|---|---|
| | Z500 [$m^2/s^2$] | T850 [K] | T2M [K] | Z500 [$m^2/s^2$] | T850 [K] | T2M [K] |
| *2 predictors* | | | | | | |
| Rasp et al. (2020) | 626 | 2.87 | - | 757 | 3.37 | - |
| Ours | 531 | 2.38 | - | 717 | 3.03 | - |
| *117 predictors* | | | | | | |
| Rasp & Thuerey$^{\text{cont}}$ | 331 | 1.87 | 1.60 | 545 | 2.57 | 2.06 |
| Rasp & Thuerey | 314 | 1.79 | 1.53 | 561 | 2.82 | 2.32 |
| Ours | 329 | 1.62 | 1.29 | 601 | 2.57 | 1.89 |

Similar patterns may appear at different positions and orientations; equivariance allows filter reuse.